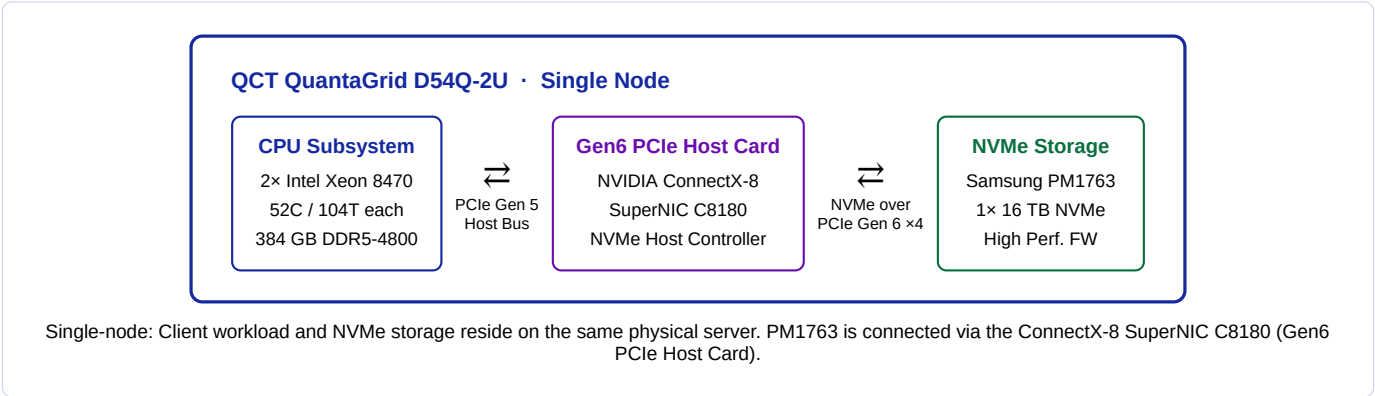


MLPerf™ Storage Benchmark v3.0

System Description — Samsung PM1763 on QCT QuantaGrid D54Q-2U

System Name	Samsung-PM1763-D54Q
Submitter	Samsung
Benchmark Version	MLPerf Storage v3.0
Submission Division	CLOSED
Solution Type	Local Storage (NVMe SSD, Direct-Attached)
Storage Access Protocol	NVMe (PCIe Gen 6)
Platform	QCT QuantaGrid D54Q-2U
NIC / Host Card	Gen6 PCIe Host Card (ConnectX-8 SuperNIC C8180)

System Topology



1. System Overview

This submission evaluates Samsung PM1763 NVMe SSD performance on the MLPerf Storage v3.0 benchmark suite (Training, Checkpointing, Vector Database, KV-Cache) using a single QCT QuantaGrid D54Q-2U server in a direct-attached local storage configuration. The PM1763 NVMe SSD is connected through the ConnectX-8 SuperNIC C8180 Gen6 PCIe Host Card, which serves as the NVMe host controller.

2. Host Node Specification

Field	Value
Platform	QCT QuantaGrid D54Q-2U
CPU	2× Intel Xeon 8470 (Sapphire Rapids), 52 Cores / 104 Threads each, 105W TDP
Total Cores / Threads	104 Cores / 208 Threads
CPU Base Frequency	2.0 GHz
CPU Max Turbo Frequency	3.8 GHz
Memory	384 GB DDR5-4800 Mbps ECC
Host Card	NVIDIA ConnectX-8 SuperNIC C8180 (Gen6 PCIe Host Card)
Storage	Samsung PM1763, 1× 16 TB NVMe SSD via PCIe Gen 6 ×4
OS	Linux (kernel version 6.11)
Node Role	Client + Storage (Single Node, Direct-Attached)

3. Software Configuration

Component	Version / Detail
MLPerf Storage Tool	mlpstorage v3.0
DLIO Benchmark	dllo-benchmark (pinned per mlpstorage lock)
Python	3.12
MPI Implementation	OpenMPI / MPICH
CUDA / Driver	N/A (storage-only benchmark)
Filesystem	xfs— local mount
Mount Options	noatime, nodiratime
I/O Scheduler	none (NVMe default)
OS Kernel	Linux (kernel version 6.11)
BIOS / Firmware Settings	Performance mode, C-states disabled, Hyper-Threading enabled

4. Network Configuration

Field	Value
Topology	Single Node — No network storage traffic
Host Card	NVIDIA ConnectX-8 SuperNIC C8180 (Gen6 PCIe)
Host Card Role	NVMe Host Controller for Samsung PM1763 (PCIe Gen 6 ×4)
Network Switches	N/A